



Evaluating NER Methods in AI Voice Onboarding on AWS

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Introduction



Context Analysis:

- Traditional onboarding forms are often slow, unintuitive, and a major cause of churn.

Objective:

- Enable voice interaction for a more natural, faster and inclusive experience (accessibility).

The Hindeep case:

- Voice onboarding for the Hindeep dating app powered by AI techniques.

Solution Approach:

- Develop a serverless prototype (AWS) for voice onboarding.
- Evaluate 3 NER strategies (Comprehend, BERT, LLM).
- Identify the optimal solution for the domain and system architecture.



Hindeep
Humans in deep

Company:

Hindeep S.r.l. (Milan).

Duration of the internship:

4 months.

Activity:

Internship within the R&D team on AI-based onboarding.

Methodology:

Agile approach with weekly sprints and task management via Trello.

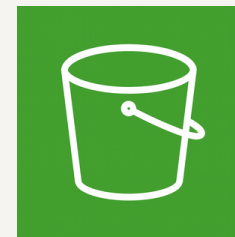
Learning outcomes:

Hands-on experience with AWS services and practical skills in serverless architecture.

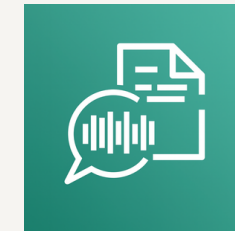
Backend System Components

Context

The system logic is implemented through AWS services, ensuring **scalability**, **cost efficiency** and **reliability**.



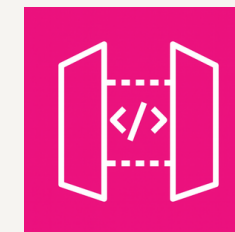
Amazon S3: used as object storage for audio files, transcriptions, and JSON profiles.



Amazon Transcribe: AI service used for speech-to-text conversion.



AWS Lambda: the “brain” of the system, responsible for code execution and process orchestration.



API Gateway: serves as a secure entry point connecting the frontend with backend functions.



Frontend System Components

Context

The frontend provides the user interaction interface, handling audio acquisition, backend communication and profile visualization.



HTML/CSS

User Interface: used to define the semantic structure and visual style of the registration and profile views.

MEDIARECORDER API

Recording Logic: used to manage microphone access, capture audio and create a local Blob for playback.

FETCH/FILEREADER API

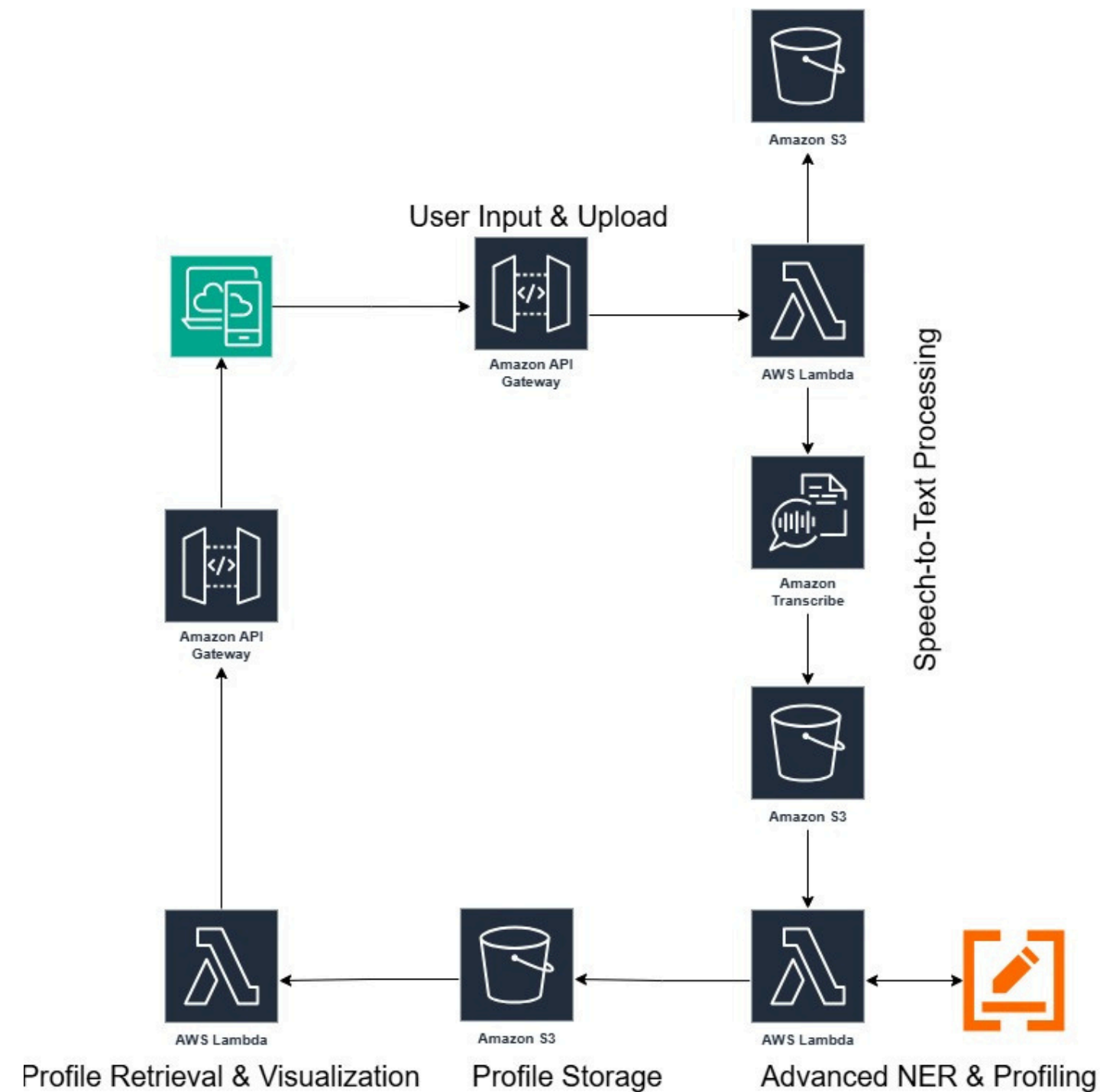
Backend Communication: used for audio conversion into Base64 and asynchronous transmission via API Gateway.

CHART.JS

Data Visualization: used for the interactive graphical representation of personality traits.



Architecture and Operational Flow



Workflow Step-by-Step

- ▶ **User Input & Upload:**
Frontend → API Gateway → Lambda 1 → S3 & Transcribe
- ▶ **Speech-to-Text Processing:**
Transcribe → S3 → trigger for Lambda 2
- ▶ **Advanced NER & Profiling:**
Lambda 2 → LLM call → stores JSON in S3
- ▶ **Profile Retrieval & Visualization:**
Frontend → API Gateway → Lambda 3 → S3 → user profile



NER

Named Entity Recognition



Named Entity Recognition (NER)

A Natural Language Processing technique that allows the automatic identification and classification of named entities in a text.



BIO Scheme:

- **B-TYPE:** Beginning of an entity.
- **I-TYPE:** Inside of an entity.
- **O:** Outside any entity.



Evaluation Metrics:

- **Precision:** Proportion of correctly identified entities among predictions.
- **Recall:** Proportion of actual entities successfully retrieved.
- **F1-Score:** Harmonic mean of precision and recall.



The Challenge of Custom NER

Corpus	Year	Text Source
MUC-6	1995	Wall Street Journal
MUC-6 Plus	1995	Additional news to MUC-6
MUC-7	1997	New York Times news
CoNLL03	2003	Reuters news
ACE	2000 - 2008	Transcripts, news
OntoNotes	2007 - 2012	Magazine, news, web, etc.
W-NUT	2015 - 2018	User-generated text
BBN	2005	Wall Street Journal
WikiGold	2009	Wikipedia
WiNER	2012	Wikipedia
WikiFiger	2012	Wikipedia
HYENA	2012	Wikipedia
N ³	2014	News
Gillick	2016	Magazine, news, web, etc.
FG-NER	2018	Various
NNE	2019	Newswire
GENIA	2004	Biology and clinical text
GENETAG	2005	MEDLINE
FSU-PRGE	2010	PubMed and MEDLINE
NCBI-Disease	2014	PubMed
BC5CDR	2015	PubMed
DFKI	2018	Business news and social media

Overview of NER corpora

Objective:

- Extract non-standard entities for the dating domain.

Entities of Interest:

- **Anagraphic:** NOME, ETA, GENERE, CITTA, ALTEZZA, PROFESSIONE.
- **Personal:** HOBBY, PERS (personality traits).
- **Relational:** INTERESSE (partner's gender), RANGEETA (partner's age range).

Problem:

- Public datasets (e.g., CoNLL03) do not contain the required entities.
- Need for a customized solution.





Solution Analysis

Amazon Comprehend

BERT

LLM Zero Shot

Solution Analysis

01 Amazon Comprehend



- ▶ **Concept:**
Use an AWS-managed NLP service that provides ready-to-use NER.
- ▶ **Pro:**
Simple integration.
- ▶ **Cons:**
 - Standard entities unsuitable.
 - Unsustainable customization (fixed endpoint cost, complexity).
- ▶ **Decision:**
Discarded.

BERT

LLM Zero Shot

Solution Analysis

02 Fine-Tuning BERT



Amazon
Comprehend

- ▶ **Concept:**
Specialize a pre-trained model (fine-tuning) for the specific task.
- ▶ **Pro:**
Promising results (average F1-Score of 0.8).
- ▶ **Cons:**
 - Model size (>1 GB) incompatible with AWS Lambda (250 MB limit).
 - Alternative (SageMaker) not cost-effective.
- ▶ **Decision:**
Discarded due to practical and cost constraints.

LLM Zero Shot



03 LLM Zero-Shot

Amazon
Comprehend

BERT

- ▶ **Concept:**
Apply prompt engineering to a general-purpose model instead of training.
- ▶ **Implementation:**
A lightweight Lambda makes an API call to OpenRouter (DeepSeek model).
- ▶ **Pro:**
 - Extreme Flexibility: on-the-fly updates by modifying the prompt.
 - Full Serverless Compatibility: meets Lambda constraints.
 - Cost: no fixed costs.
- ▶ **Cons:**
 - Dependence on external APIs.
 - Non-deterministic output.

Final Choice: LLM Zero-Shot

Critical Comparison

*"Mi chiamo Marco, ho 32 anni e vivo a Bologna.
Lavoro come insegnante di storia in un liceo,
ma in passato ho fatto l'archeologo in Grecia per alcuni anni.
Sono una persona istintiva e flessibile ma non altruista,
mi piace cucinare e suonare la chitarra.
Cerco una donna tra i 30 e i 34 anni."*

02 Fine-Tuning BERT

```
{  
  "NOME": "Marco",  
  "ETA": "32",  
  "CITTA": "Bologna.",  
  "PROFESSIONE": ["insegnante", "storia", "archeologo"],  
  "PERS": "istintiva e flessibile ma non altruista",  
  "HOBBY": ["cucinare", "suonare", "chitarra."],  
  "INTERESSE": "donna",  
  "RANGEETA": "30 e i 34"  
}
```

VS

03 LLM Zero-Shot

```
{  
  "NOME": "Marco",  
  "ETA": "32", "GENERE": "Uomo",  
  "CITTA": "Bologna",  
  "PROFESSIONE": "insegnante di storia",  
  "PERS": ["Flessibile", "Istintiva"],  
  "HOBBY": ["cucinare", "suonare la chitarra"],  
  "INTERESSE": "donna",  
  "RANGEETA": {"min": "30", "max": "34"}  
}
```

Conclusions

- ▶ **Achieved Objective :**
Validated the feasibility of a voice-based, serverless, AI-powered onboarding system.
- ▶ **Winning Choice:**
The Zero-Shot LLM approach proved to be the most pragmatic, flexible and effective.

Future Developments

- ▶ **Conversational Onboarding** (interactive dialogue).
- ▶ **Emotional Voice Analysis.**
- ▶ **Integration with Serverless Databases (DynamoDB).**
- ▶ **Applications in Other Domains**
(HR, Customer Service, Healthcare).

Conclusions & Future Developments





Thank You



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Corpus	Year	Text Source	#Tags	URL
MUC-6	1995	Wall Street Journal	7	https://catalog ldc.upenn.edu/LDC2003T13
MUC-6 Plus	1995	Additional news to MUC-6	7	https://catalog ldc.upenn.edu/LDC96T10
MUC-7	1997	New York Times news	7	https://catalog ldc.upenn.edu/LDC2001T02
CoNLL03	2003	Reuters news	4	https://www.clips.uantwerpen.be/conll2003/ner/
ACE	2000 - 2008	Transcripts, news	7	https://www ldc.upenn.edu/collaborations/past-projects/ace
OntoNotes	2007 - 2012	Magazine, news, web, etc.	18	https://catalog ldc.upenn.edu/LDC2013T19
W-NUT	2015 - 2018	User-generated text	6/10	http://noisy-text.github.io
BBN	2005	Wall Street Journal	64	https://catalog ldc.upenn.edu/LDC2005T33
WikiGold	2009	Wikipedia	4	https://figshare.com/articles/Learning_multilingual_named_entity_recognition_from_Wikipedia/5462500
WiNER	2012	Wikipedia	4	http://rali.iro.umontreal.ca/rali/en/winer-wikipedia-for-ner
WikiFiger	2012	Wikipedia	112	https://github.com/xiaoling/figer
HYENA	2012	Wikipedia	505	https://www.mpi-inf.mpg.de/departments/databases-and-information-systems/research/yago-naga/hyena
N ³	2014	News	3	http://aksw.org/Projects/N3NERNEDNIF.html
Gillick	2016	Magazine, news, web, etc.	89	https://arxiv.org/e-print/1412.1820v2
FG-NER	2018	Various	200	https://fgner.alt.ai/
NNE	2019	News wire	114	https://github.com/nickyringland/nested_named_entities
GENIA	2004	Biology and clinical text	36	http://www.geniaproject.org/home
GENETAG	2005	MEDLINE	2	https://sourceforge.net/projects/bioc/files/
FSU-PRGE	2010	PubMed and MEDLINE	5	https://julielab.de/Resources/FSU_PRGE.html
NCBI-Disease	2014	PubMed	1	https://www.ncbi.nlm.nih.gov/CBBresearch/Dogan/DISEASE/
BC5CDR	2015	PubMed	3	http://bioc.sourceforge.net/
DFKI	2018	Business news and social media	7	https://dfki-lt-re-group.bitbucket.io/product-corpus/